



SPECIAL TOPIC ARTICLE

CAReFL: Enhancing smart healthcare with Contribution-Aware Federated Learning

Zelei Liu^{1,2} | Yuanyuan Chen¹ | Yansong Zhao¹ | Han Yu¹ | Yang Liu³ | Renyi Bao⁴ | Jinpeng Jiang⁴ | Zaiqing Nie³ | Qian Xu⁵ | Qiang Yang^{5,6}

¹School of Computer Science and Engineering, Nanyang Technological University, Singapore

²Unicom (Shanghai) Industrial Internet Co. Ltd., Shanghai, China

³Institute for AI Industry Research, Tsinghua University, Beijing, China

⁴Yidu Cloud Technology Inc., Beijing, China

⁵Department of AI, WeBank, Shenzhen, China

⁶Department of Computer Science and Engineering, Hong Kong University of Science and Technology, Hong Kong

Correspondence

Han Yu, School of Computer Science and Engineering, Nanyang Technological University, Singapore.

Email: han.yu@ntu.edu.sg

Yang Liu, Institute for AI Industry Research, Tsinghua University, Beijing, China.

Email: liuy03@air.tsinghua.edu.cn

Jinpeng Jiang, Yidu Cloud Technology Inc., Beijing, China.

Email: jinpeng.jiang@yiduccloud.cn

Qiang Yang, Department of AI, WeBank, Shenzhen, China.

Email: qyang@cse.ust.hk

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Abstract

Artificial intelligence (AI) is a promising technology to transform the healthcare industry. Due to the highly sensitive nature of patient data, federated learning (FL) is often leveraged to build models for smart healthcare applications. Existing deployed FL frameworks cannot address the key issues of varying data quality and heterogeneous data distributions across multiple institutions in this sector. In this paper, we report our experience developing and deploying the Contribution-Aware Federated Learning (CAReFL) framework for smart healthcare. It provides an efficient and accurate approach to fairly evaluate FL participants' contributions to model performance without exposing their private data, and improves the FL model training protocol by allowing the best performing intermediate sub-models to be distributed to participants for FL training. Since its deployment by Yidu Cloud Technology Inc. in March 2021, CAReFL has served eight well-established medical institutions in China to build healthcare decision support models. It can perform contribution evaluations close to three times faster than the best existing approach and has improved the average accuracy of the resulting models by more than 2% compared to the previous system (which is significant in industrial settings). To the best of our knowledge, it is the first CAReFL successfully deployed in the healthcare industry.

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INTRODUCTION

Artificial intelligence (AI) technologies are becoming ubiquitous in our daily lives. For machine learning-based approaches, having access to high-quality data is crucial for achieving good performance. This is especially true for AI-empowered smart healthcare applications. However, medical data owned by a single entity (e.g., a hospital, a research institution) are often not enough to achieve this goal. Hence, collaborative model training is necessary for this field (Warnat-Herresthal et al. 2021).

Medical data are highly sensitive, making direct data sharing among healthcare institutions challenging. Recent data privacy protection laws around the world, such as the General Data Protection Regulation (GDPR) (GDPR 2018), even more unattainable. The concern for medical data privacy protection has long been existing and regulated, which makes it hard to develop AI solutions the traditional way by directly collecting data from multiple sources. To enable AI to continue advance under such constraints, the field of federated learning (FL) (Kairouz et al. 2021; Yang et al. 2020), an emerging privacy-preserving collaborative machine learning paradigm, has witnessed rapid development in recent years. It has been applied in fields from safety management (Liu et al. 2020), banking (Long et al. 2020) to industry 4.0 (Chen et al. 2023). FL has been adopted by applications in different scales, from edge devices to enterprise level collaboration (Yang et al. 2020).

FL-empowered smart healthcare applications have emerged in recent years (Kaissis et al. 2020; Rieke et al. 2020; Sadilek et al. 2021; Sheller et al. 2020; Xu et al. 2020). They are generally built based on the popular Federated Averaging (FedAvg) FL model training approach (McMahan et al. 2017). While such frameworks can support privacy-preserving collaborative model training, they are not well suited for situations involving data heterogeneity due to non-i.i.d. statistical data distributions across the data silos, which is common among different healthcare institutions. In addition, raw healthcare data are often scattered, unstructured, and non-standardized. Thus, data quality across multiple healthcare institutions may vary. Currently, they need to be processed by professionals in order to produce useful FL models.

In addition to data heterogeneity and data quality concerns, stakeholders involved in a healthcare ecosystem may need to attribute contributions to individual data owners. For example, a pharmaceutical company may wish to build a model to facilitate drug research by leveraging data from multiple hospitals through FL. In order to compensate the participating hospitals, the pharmaceutical company may need to offer incentive payouts. However, without being able to directly assess the quality

of each hospital's local data, fairly compensating them can be challenging (Lyu et al. 2020).

To address these issues, we present the Contribution-Aware Federated Learning (CAreFL) framework in this paper. Built on top of the GTG-Shapley FL client contribution evaluation (CE) method we have previously developed (Liu et al. 2022), it provides fair and accurate FL participant CE efficiently without the need to directly inspect the original data. In addition, it leverages the knowledge about the performance of various sub-models generated during Shapley value (SV) estimation to offer a new “survival of the fittest” FL model aggregation approach. Last but not least, the framework keeps track of the historical CE records and converts them into reputation values for the FL participants. This information serves as a basis for managing incentive allocation, selecting future FL participants, or helping targeted healthcare institutions find ways to resolve any data quality issues they may face.

Since March 2021, CAreFL has been deployed by WeBank¹ and Yidu Cloud Technology Inc.² to support FL model training involving eight well established healthcare institutions in China. The deployment results show that it can perform CEs 2.84 times faster than the best existing approach. Compared to the previous FedAvg-based FL model training approach used by Yidu Cloud, CAreFL achieved a 2.62% model accuracy improvement on average, which is significant in industrial smart healthcare applications. To the best of our knowledge, it is the first CAreFL successfully deployed in the healthcare industry.

SYSTEM DESIGN

The industry partner deploying CAreFL – Yidu Cloud Inc. – offers healthcare solutions built on big data and AI technologies. It serves hospitals, pharmaceutical companies, biotech and medical device companies, research institutions, insurance companies, as well as regulators and policy makers. Based on its independently developed Data Process and Application Platform, Yidu Cloud is capable of processing mass multi-source heterogeneous healthcare data. It has built up a platform to support multi-center secure computing based on the FL architecture to enable multi-party collaborative AI solution development (Figure 1). The platform has been certified by the National Institute of Information and Communication. At present, the platform supports multiple FL models trained with the FedAvg approach. Through this research work, we incorporate CAreFL into the platform as an alternative FL model training approach to FedAvg.

In this section, we provide detailed descriptions of the CAreFL framework. It consists of three tiers as shown in Figure 1. They are: (1) the FL infrastructure tier, which

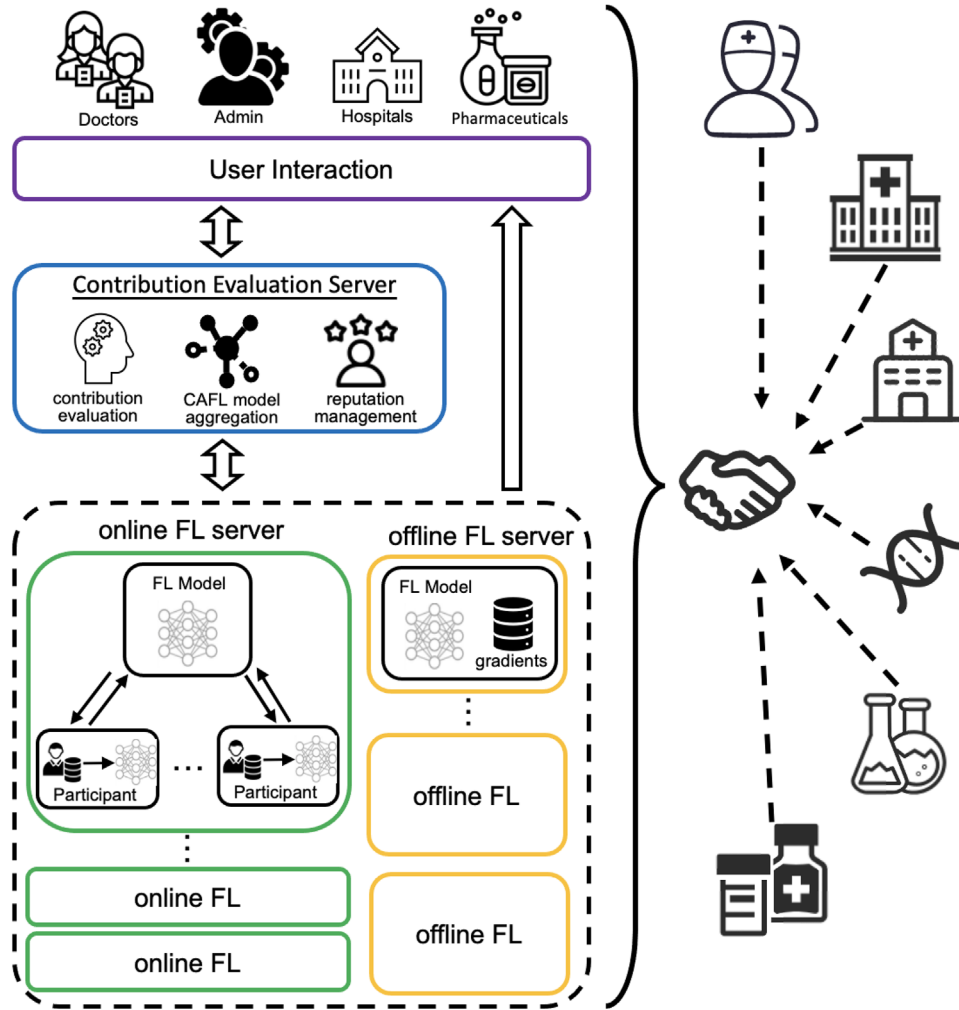


FIGURE 1 An overview of the Yidu Cloud federated learning (FL)-based multi-center collaborative model training platform.

consists of the online FL server (currently training an FL model) and the offline FL server (storing previously trained models), (2) the CE tier, and (3) the user interaction tier.

Contribution evaluation

The CE workflow of CAReFL is illustrated in Figure 2. It can be performed in either the online mode or the offline mode depending on the users' preference. The online mode evaluates participants' contributions during the FL model training process, whereas the offline mode evaluates their contributions for FL models which have already been built (this would require historical local model updates being recorded by the system). Both modes use the same CE algorithm – GTG-Shapley (Liu et al. 2022).

The CAReFL CE process in each round of FL model training is as follows:

- Step 1: The FL server initiates an FL session by sending a homomorphic encryption (HE) key to the FL participants.
- Step 2 and 3: The FL server performs FL model training as per normal. This process is entirely separated from the CAReFL. Thus, the adoption of CAReFL does not affect how FL model training is performed by the original system.
- Step 4: The FL server sends an evaluation request to the CAReFL CE server following a predefined message format.
- Step 5: The FL server sends the obfuscated IDs of FL participants joining the current round of training to the CAReFL CE server.
- Step 6: The FL server reconstructs an FL sub-model with only the model updates from the set of participants identified by the CAReFL CE server. Then, it evaluates the performance of the sub-model and sends the results back to the CE server. The CE server carries out

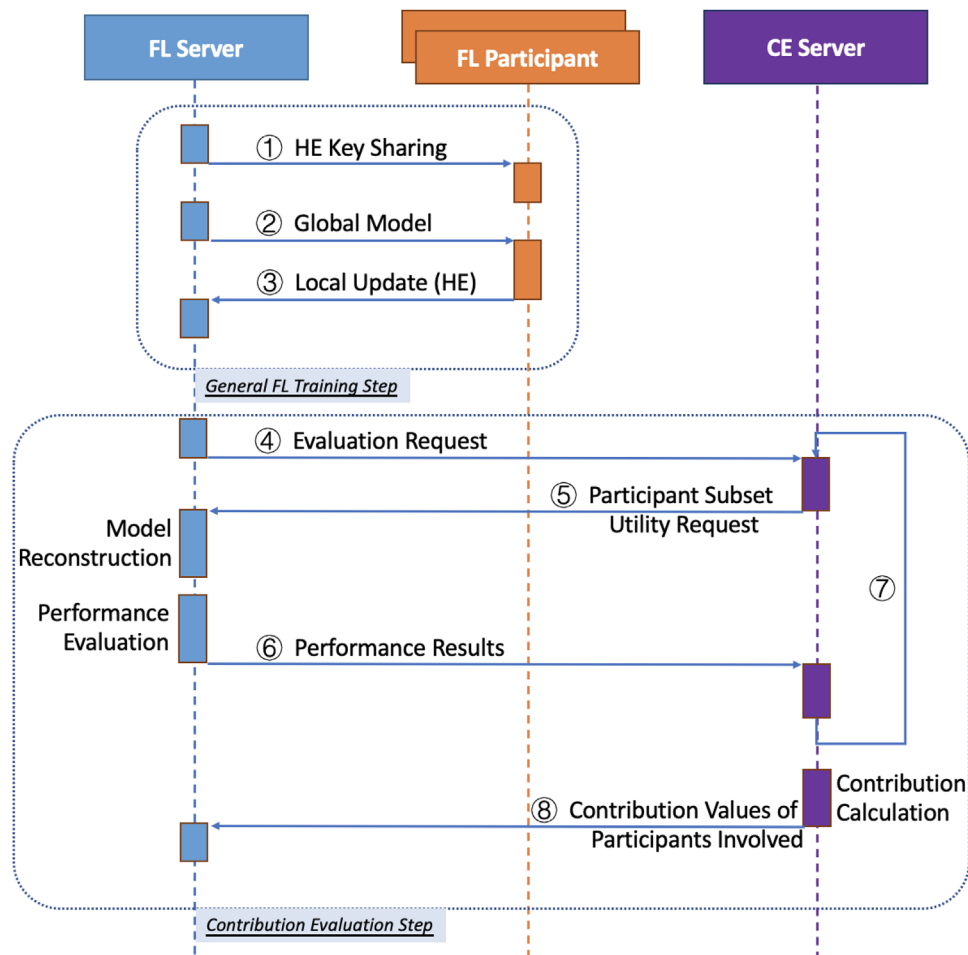


FIGURE 2 The Contribution-Aware Federated Learning (CAREFL) contribution evaluation workflow and how it interacts with the horizontal federated learning model training process.

SV-based CE by leveraging the GTG-Shapley algorithm. Once this is complete, it sends the set of participant IDs selected by the proposed GTG-Shapley algorithm to the FL server to perform sub-model training and evaluation.

- Step 7: The CE server analyzes the received results and adjusts the selected participants set. The new set of participants are sent to the FL server to be evaluated.
- Step 8: When the CE server determines that enough information has been gathered, it stops further sub-model evaluation. The contribution values for each participant are sent to FL the server.

Here, we provide an overview of how GTG-Shapley works. When evaluating an FL participant's contribution, direct access to its local data is now allowed to avoid exposing potentially sensitive information. Therefore, the CE process must be carried out without examining the raw data. SV (Shapley 1953) is a classic approach which has been shown to fairly quantify the contributions of individuals within a coalition. Moreover, it only requires the final utility achieved by the coalition for calculation. These

properties make it a frequently adopted method for FL participant CE.

Computing the canonical SV takes exponential time with regard to the number of FL participants involved. In addition, training FL models on large local data silos can be time consuming, too. Furthermore, the sizes hence, the canonical SV is rarely directly used for CE in FL. Several approaches for accelerating SV calculation in the context of FL have been proposed. (Ghorbani and Zou 2019) samples part of the entire set of possible participant combinations to reduce SV calculation complexity. However, as it still required retraining of the subset FL models, it is still too computationally expensive. Therefore, gradient-based FL model reconstruction approaches (Song, Tong, and Wei 2019; Wang et al. 2020) have been proposed to avoid FL sub-model retraining. These approaches do relieve a large proportion of the SV computational overhead. However, they still require $O(2^N)$ FL sub-model reconstructions, where N is the number of FL participants collaboratively training a model. Thus, they cannot be scaled up to large federations. The key idea of GTG-Shapley is to

opportunistically reduce the need for sub-model retraining with model reconstruction, while strategically sampling different combinations of FL participants in order to produce an accurate estimation of SV efficiently.

Elimination of sub-model retraining: GTG-Shapley leverages the performance information of reconstructed FL models to generate the necessary sub-models. The utility produced by a subset of FL participants represents their collective contribution to the FL model, which stem from their individual model gradient updates. Thus, once GTG-Shapley determines that the utility of a given subset of participants needs to be evaluated, the FL server reconstructs the sub-model involving these participants based on their model updates it has received, instead of retraining from scratch.

Guided truncation of model evaluations: Apart from sub-model retraining, the majority of the time required to estimate SVs is due to the exponential number of FL sub-model reconstruction and evaluation. In addition, FL models with a large number of parameters or federations with large test datasets can also significantly slow down SV estimation. To address these issues, GTG-Shapley strategically prunes unnecessary sub-models. This is achieved through the following two techniques.

Firstly, an FL participant's SV is estimated based on its marginal gain in different subsets, and the FL server performs SV utility evaluation. FL models tend to show patterns of diminishing returns both on the marginal gains of within each round of evaluation orders in a sampled permutation, as well as the marginal gains across different rounds. A participant can be categorized as "valuable" or "not valuable" based on its marginal gain. GTG-Shapley only evaluates the "valuable" FL participants in a sampled permutation. The rest can be pruned without significantly affecting SV estimation. Secondly, the marginal utility gains tend to concentrate at the leading positions in a given FL participant permutation. Thus, it is important to ensure that participants have equitable opportunities to occupy different positions across multiple permutations in order to evaluate their contributions fairly.

To take advantage of these insights, GTG-Shapley is incorporated with a between-round truncation policy to opportunistically skip entire rounds of SV calculation when the remaining marginal utility gain is deemed to be insignificant. In addition, a within-round truncation policy is put in place for opportunistically terminating the ongoing sub-model evaluation when the marginal gain of the remaining FL participants in the current permutation is deemed to be insignificant. Lastly, GTG-Shapley has a guided sampling policy to fairly place FL participants in different positions across multiple participant subsets.

Best-performing sub-model propagation

During the CE process, CAReFL assesses and keeps track of the performance of the FL sub-models. For instance, as shown in Figure 3 under the Top Models panel, the performance of the evaluated FL sub-models in a given round of training are ranked in descending order from top to bottom in each column. The ones marked with a dot denotes the performance of the full FL model (involving all clients). It can be observed that the full FL models do not always achieve the best performance. This enables the CE server to identify the sub-model with the best performance and leverage this information to guide the FL server to performance model updating across multiple training rounds.

Thus, after computing the FL participant contribution values for a training round, the CE server returns an additional subset of participants whose local model updates produce the best performing aggregated FL model. This subset can be viewed as the "best subset" of participants for FL model aggregation. With this information, the FL server reconstructs an FL model with the local model updates only from participants in the "best subset." This "best subset" FL model is then distributed to all the participants as the new global FL model for the next round of collaborative training (Figure 4).

Participant reputation update

The FL server often needs to select a number of participants from the candidate pool when initiating a new FL model training task. To facilitate this process, CAReFL keeps track of the participants' track records during past CE rounds to build reputation profiles for FL participants.

The CE track records can be used in a wide range of reputation evaluation models to produce reputation values for FL participants. In the current implementation, we adopt well-established principles of reputation evaluation from the multi-agent systems literature (Shen et al. 2011; Yu et al. 2013), emphasizing on context-awareness and recency. For example, a participant's data might be more valuable for building an FL model for cancer research than one for bone fracture identification. Moreover, as a participant's data quality depends on its data collection and processing effort, its perceived contribution value in the same application context may also change over time.

The CAReFL platform adopts the Beta Reputation System (BRS) (Josang and Ismail 2002). Based on FL participants' contribution records, it discretizes each record into "good" and "bad" to make it suitable for BRS. If a participant's contribution value is higher than the

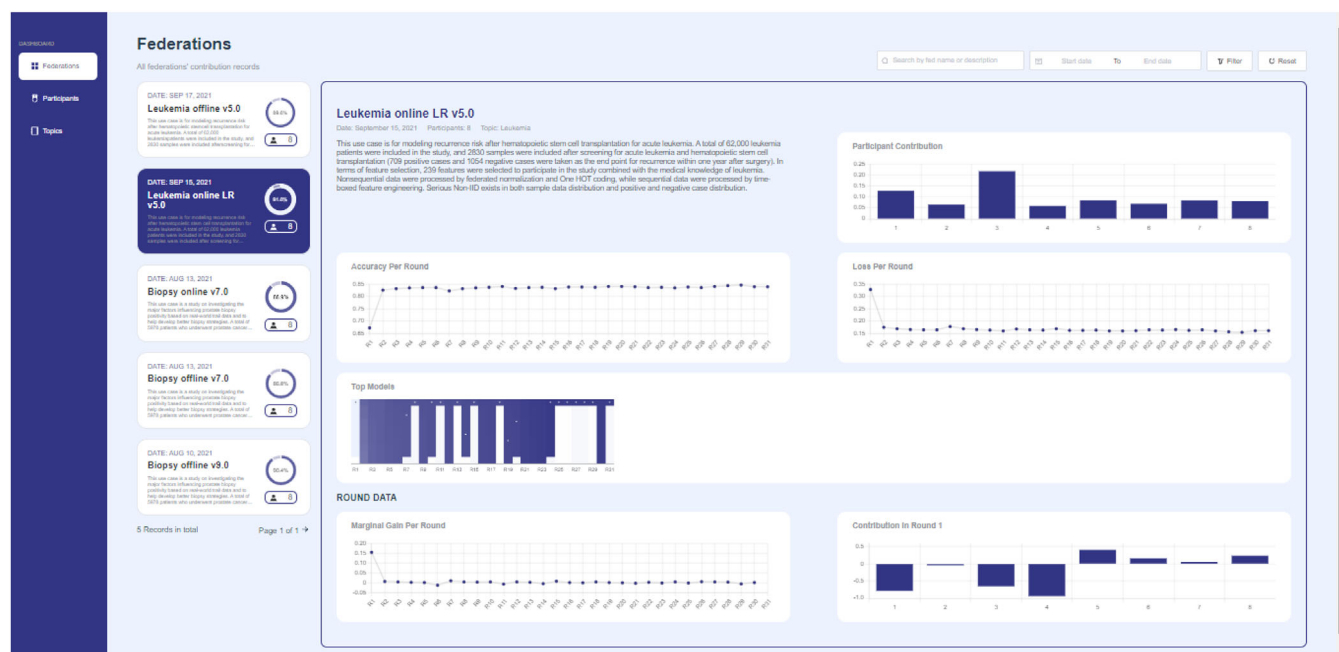


FIGURE 3 The Contribution-Aware Federated Learning (CAREFL) system control panel for federated learning (FL) model training process³.

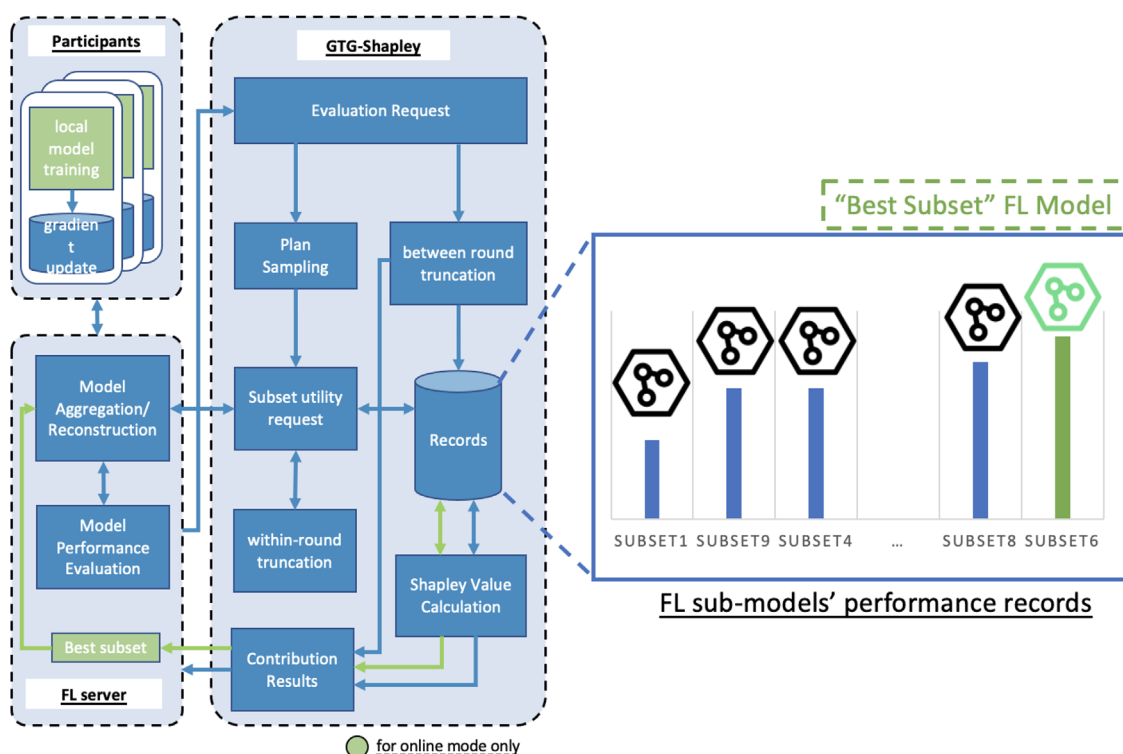


FIGURE 4 The Contribution-Aware Federated Learning (CAREFL) artificial intelligence (AI) engine.

average contribution value of all the participants in this round, this record is categorized as “good.” Otherwise, it is categorized as “bad.” CAREFL calculates every participants’ reputation under each context based on BRS. A participant’s overall reputation is the average of their

reputation values over all the contexts they have joined before. The reputation scores for each FL participant are organized according to the application context and the time stamp to facilitate reputation evaluation as shown in Figure 5.

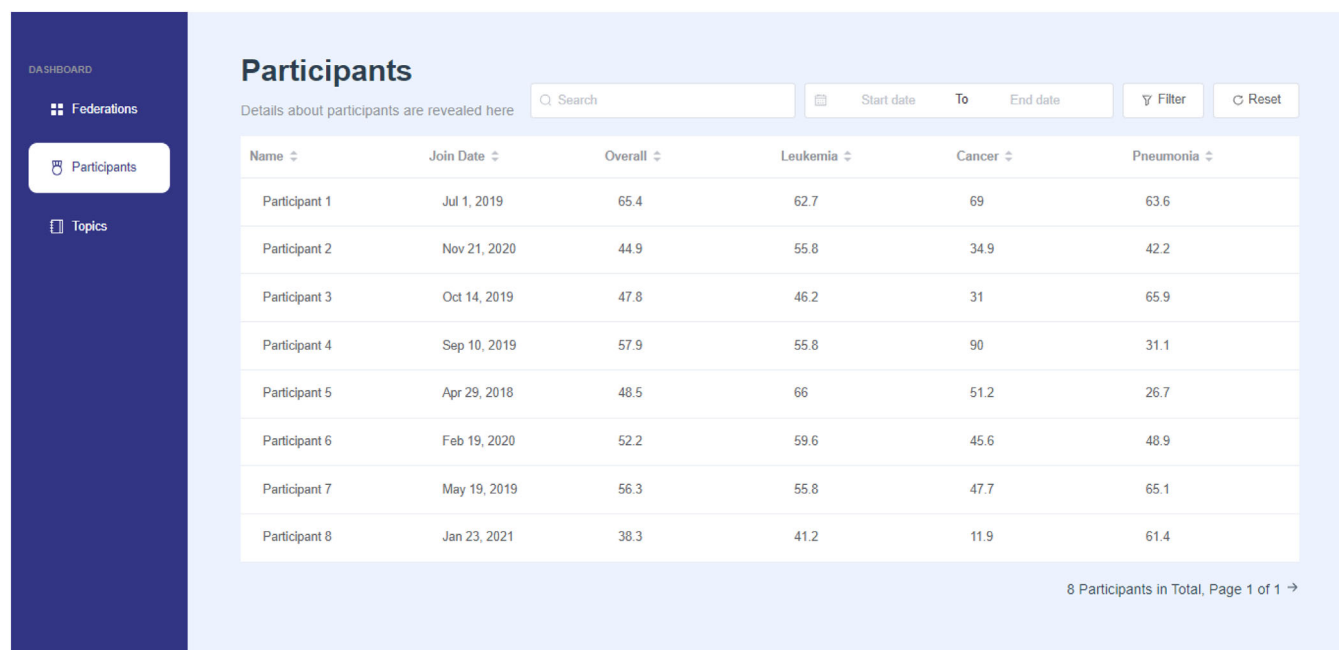


FIGURE 5 The Contribution-Aware Federated Learning (CAreFL) participant reputation dashboard⁴.

IMPACT

The CAreFL framework has been deployed in Yidu Cloud Technology Inc. since March 2021 in two lines of their business: (1) clinical research services, and (2) real-world trial research services. Clinical research focuses on training FL models involving data silos from multiple hospitals. Real world trial research is often initiated by a pharmaceutical company which aims to leverage data from multiple hospitals to build models. Both services require data which need to be collected by the hospitals over months or years under their respective Institutional Review Board (IRB) supervision. So far, CAreFL has been used to help eight well-known medical institutions in China to train AI models for risk prediction, disease diagnosis, and influence factor analysis.

Leukemia: CAreFL has been used to train models for recurrence risk prediction after hematopoietic stem cell transplantation for acute leukemia treatment. This is a clinical research business case. A total of 62,000 patients were included in the study, and 2830 samples are included after acute leukemia screening and hematopoietic stem cell transplantation (709 positive cases and 1054 negative cases at the end point of the recurrence study 1 year after the surgery). The results for this case are shown in Figure 6A,B. A logistic regression (LR) model and a homo SecureBoost Tree (SBT) model (Cheng et al. 2021) have been trained with FedAvg in the previous system, and later retrained with CAreFL after its deployment. In both cases, CAreFL quickly reaches the performance plateau. It outperforms the previous system (with FedAvg)

by 3.34% and 2.83% for the LR model and the SBT model, respectively.

Biopsy: CAreFL has been used to train models for analyzing major factors influencing prostate biopsy positivity based on real-world trial data. This is a clinical research business case. A total of 5978 patients who underwent prostate cancer biopsy during a 5-year period were screened, and 2426 patients are selected. The model mainly analyzes the relationship between patient age, family history, tPSA/f-PSA, PSAD, preoperative testosterone, preoperative MRI, preoperative DRE, Gleason score, and other factors with positive biopsy diagnosis. The results for this case are shown in Figure 7A,B. CAreFL outperforms the previous system by 3.41% and 2.22% for the LR model and the SBT model, respectively.

Pneumonia: CAreFL has also been used to train models for classifying whether hospitalized pneumonia patients will be transferred to the intensive care unit (ICU) or pass away. This is a real-world trial research business case. A total of 103,455 sample data were selected based on hospitalization, death, and transfer to ICU for continued treatment during hospitalization. These samples contain 57 features, including basic patient information, vital signs, test results, symptoms, and other information. The results for this case are shown in Figure 8A,B. CAreFL outperforms the previous system by 2.87% and 1.00% for the LR model and the SBT model, respectively.

With the help of CAreFL, Yidu Cloud has avoided the problems of high computation costs and low accuracy of existing FL participant CE approaches. This capability not only allows it to provide more detailed analysis

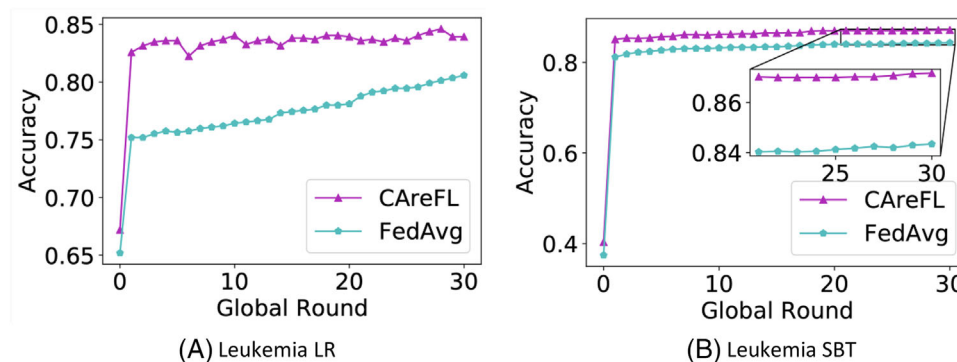


FIGURE 6 Deployment comparison results for Leukemia.

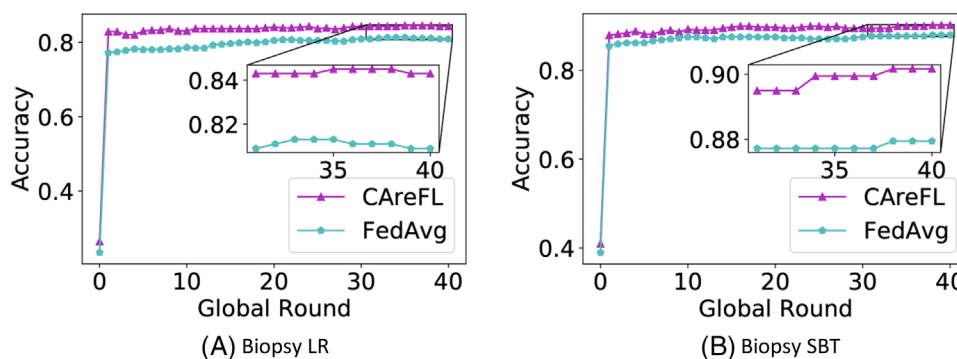


FIGURE 7 Deployment comparison results for Biopsy.

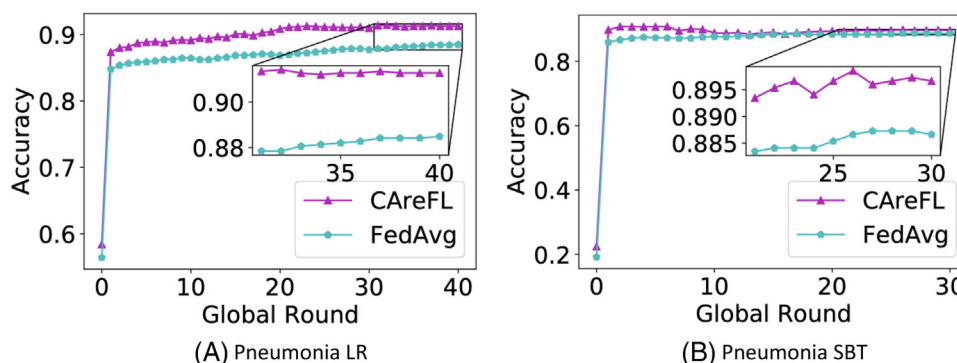


FIGURE 8 Deployment comparison results for Pneumonia.

about the value each FL participant can bring into building any given FL model without exposing their sensitive data, but also enables a new CAreFL model aggregation approach to be developed. Overall, CAreFL is 2.84 times faster than the best performing baseline, MR, in terms of evaluating participants' contributions. It has improved the average accuracy of FL models trained by Yidu Cloud by 2.62% compared to its previous system, which is significant in industrial smart healthcare settings.

CONCLUSIONS AND FUTURE WORK

In this paper, we reported on our experience using CAreFL to enhance privacy-preserving collaborative training of machine learning models involving multiple healthcare industry data owners. We developed the CAreFL framework which provides fair and explainable FL participant CE in an efficient and privacy-preserving manner, and optimizes the FL model aggregation approach based on the evaluation results. Since its deployment in March 2021

in Yidu Cloud, CAreFL has helped eight well-established medical institutions in China to train machine learning models for healthcare decision support, and has made significant positive impact by improving CE efficiency and model performance compared to the previous system. To the best of our knowledge, it is the first successfully deployed CAreFL framework in the healthcare industry.

In future, we will continue to explore the applicability of CAreFL in other smart healthcare application scenarios. We will also extend the CAreFL framework with contribution-based data pricing mechanisms (Pei 2020) to support the emergence of an FL-based healthcare data exchange marketplace. Eventually, we aim to incorporate these functionalities into the open-source FATE framework and make them available to more developers, researchers, and practitioners.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict.

ORCID

Han Yu  <https://orcid.org/0000-0001-6893-8650>

ENDNOTES

¹<https://www.webank.com/>

²<https://www.yiducloud.com.cn/en/>

³<https://demo.federated-learning.org/>

⁴<https://demo.federated-learning.org/Participants>

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AUTHOR BIOGRAPHIES



Zelei Liu currently serves as the Technical Director of Unicom (Shanghai) Industrial Internet Co. Ltd. Before joining Unicom, he was a Research Fellow in School of Computer Science and Engineering, Nanyang Technological University, Singapore from 2019 to 2022. He received his PhD in Computer Science and Technology from Jilin University, China. His current research focus on federated learning and incentive mechanism design.



Yuanyuan Chen is a Research Engineer and a PhD student in the School of Computer Science and Engineering (SCSE), Nanyang Technological University (NTU), Singapore. His research interests include federated learning, and statistical learning theory, etc.



Yansong Zhao is a Research Associate and a PhD student in the School of Computer Science and Engineering (SCSE), Nanyang Technological University (NTU), Singapore. His research interests include federated learning, and collaborative federated learning

ecosystem, etc.



Han Yu is a Nanyang Assistant Professor (NAP) in the School of Computer Science and Engineering (SCSE), Nanyang Technological University (NTU), Singapore. He held the prestigious Lee Kuan Yew Post-Doctoral Fellowship (LKY PDF) from 2015 to 2018.

He obtained his PhD from the School of Computer Science and Engineering, NTU. His research focuses on federated learning and algorithmic fairness. He has published over 200 research papers and book chapters in leading international conferences and journals. He is a co-author of the book *Federated Learning* – the first monograph on the topic of federated learning. His research works have won multiple awards from conferences and journals. He is a *Senior Member* of AAAI, CCF and IEEE.



Yang Liu is an Associate Researcher and Associate Professor at the Institute for AI Industry Research, Tsinghua University. She graduated from Tsinghua University with a B.S. degree in chemical engineering, and from Princeton University

with a PhD in chemical and biological engineering. Before joining Tsinghua, she was a Principal Researcher and Research Team Lead at WeBank. She also previously worked at Air Products and Dataminr Inc. She holds more than 20 patents and more than 100 patent applications. Her research was published in well-known journals including *Nature*, *AAAI*, *IJCAI*, *USENIX*, and *ACM TIST*, and cited for more

than 10,000 times. She co-authored the first book on federated learning, and Federated Learning. She also serves as an associate editor for ACM TIST, and a guest editor for IEEE Intelligent Systems and IEEE BigData, and co-chaired multiple workshops at IJCAI and NeurIPS. Her research work has been recognized with multiple awards, such as CCF Technology Award and AAAI Innovation Award.



Renyi Bao currently serves as a server-side architect at ByteDance Inc., and has over 7 years of experience in distributed computing architectures. Before that, he worked as an architect at Yidu Cloud, where he established a secure computing team and retried

the research and product development work in multi-party computing, federated learning, blockchain, and other directions. He works on providing privacy computing solutions for medical scenarios. He has extensive engineering and research experience in distributed storage and computing.



Jinpeng (Jasper) Jiang currently serves as the Head of Technology at EYD, and has over 10 years of experience in big data and AI architecture. Previously Chief Architect of Yidu Cloud, Jinpeng was responsible for the development of medical big data infras-

tructure and platform product development. Before joining Yidu Cloud, he served as the Principal Architect of Baidu, responsible for the big data processing and online searching architecture for hundreds of billions of web pages. He has extensive experience in search engine, large-scale distributed system architecture, and medical data processing and applications. Jinpeng received his bachelor's and master's degrees in Electronic Engineering from Tsinghua University.



Zaiqing Nie is a Named Professor of Tsinghua University and Principal Researcher of AIR. He graduated with a doctoral degree from Arizona State University in 2004, where his supervisor was Subbarao Kambhampati, past President of AAAI. He was previously a gradu-

ate from Department of Computer Science and Technology, Tsinghua University. He joined Alibaba Group in October 2017 as Director of AI Lab Beijing R&D

Center and Chief Scientist behind Tmall Genie, and he was also on AI Expert Panel under the Ministry of Education. Under Nie's leadership, the AI team behind Tmall Genie innovated key algorithms of the voice assistant through far-field voice recognition, natural language understanding, personalized recommendation, etc., revolutionizing personal assistant services of the AIoT era. Prior to working at Alibaba, Dr. Nie was the Principal Researcher at Microsoft Research Asia, in charge of the R&D of natural language understanding and entity mining. He also initiated and developed the Microsoft Academic, Human Resources Cube, and the enterprise assistant EDI, and served as the technological head of LUIS, Microsoft's natural language understanding platform.



Qian Xu graduated from Hong Kong University of Science and Technology (HKUST) in 2011. She is currently an Adjunct Associate Professor of Artificial Intelligence Thrust at HKUST (GZ). Her research interests include Artificial Intelligence,

Transfer Learning, Federated Learning, Natural Language Processing, and so on. She has published many research papers and patents. She has been working for several High-tech companies for many years, including Baidu Inc. and Webank Inc. Her research results have been applied to search engine, recommendation system, and robot Xiaodu. These well-known Internet products are used by billions of users. She is Executive Director of Guangdong Society of Artificial Intelligence and Robotics, and one of founding members of Guangdong-Hong Kong-Macao Greater Bay Area Society of Artificial Intelligence and Robotics.



Qiang Yang is the head of the AI department at WeBank (Chief AI Officer) and Chair Professor at the Computer Science and Engineering (CSE) Department of the Hong Kong University of Science and Technology (HKUST), where he was a former head of CSE

Department and founding director of the Big Data Institute (2015–2018). His research interests include AI, machine learning, and data mining, especially in transfer learning, automated planning, federated learning, and case-based reasoning. He is a fellow of several international societies, including ACM, AAAI, IEEE, IAPR, and AAAS. He received his PhD in Computer Science



in 1989 and his MSc in Astrophysics in 1985, both from the University of Maryland, College Park. He obtained his BSc in Astrophysics from Peking University in 1982. He had been a faculty member at the University of Waterloo (1989–1995) and Simon Fraser University (1995–2001). He was the founding Editor-in-Chief of the ACM Transactions on Intelligent Systems and Technology (ACM TIST) and IEEE Transactions on Big Data (IEEE TBD). He served as the President of International Joint Conference on AI (IJCAI, 2017–2019) and an executive council member of Association for the

Advancement of AI (AAAI, 2016–2020). Qiang Yang is a recipient of several awards, including the 2004/2005 ACM KDDCUP Championship, the ACM SIGKDD Distinguished Service Award (2017), and AAAI Innovative AI Applications Award (2016). He was the founding director of Huawei's Noah's Ark Lab (2012–2014) and a co-founder of 4Paradigm Corp, an AI platform company. He is an author of several books including *Intelligent Planning* (Springer), *Crafting Your Research Future* (Morgan & Claypool), and *Constraint-based Design Recovery for Software Engineering* (Springer).